Chirag Lakhani

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Research Summary

I develop statistical and deep learning methods for predicting the functional effects of regulatory genetic variants, and apply them to identify causal variants and uncover missing regulation in Alzheimer's disease. My work integrates genomic deep learning models with functional genomics data to understand cell-type-specific gene regulation in neurodegenerative disease.

Positions Held

Senior Research Scientist, New York Genome Center 01/2021 – Present

- Develop deep learning methods for predicting regulatory variant effects and apply them to fine-mapping and heritability analysis in Alzheimer's disease, in collaboration with David Knowles (Columbia University) and Towfique Raj (Icahn School of Medicine at Mount Sinai).
- Lead author on two methods papers integrating genomic deep learning models with functional genomics: functionally informed fine-mapping of causal AD variants, and cell-type-resolved eQTL prediction (scEEMS) explaining AD heritability.
- Collaborate with Gao Wang (Columbia University) on molecular QTL prediction within the ADSP Functional Genomics Consortium (FunGen-AD); funded as a researcher on NIH/NIA R01 AG086467 (*FunGen-xQTL*, PI: Gao Wang) and previously on NIH/NIA U01 AG068880 (*Learning the Regulatory Code of Alzheimer's Disease Genomes*).
- Present regularly to the ADSP AI/ML Working Group and its External Advisory Board.
- Co-supervised Clara Yu (research intern, Caltech) on interpretable deep learning for chromatin QTL fine-mapping, resulting in a peer-reviewed conference paper (IEEE ICBCB 2026).
- Mentored Giacomo Cavalca (graduate student) and Rohan Nidumbur (undergraduate student), whose work contributed to the scEEMS preprint, on which both are co-authors.

Postdoctoral Research Fellow, Harvard Medical School - Department of Biomedical Informatics 03/2015 – 12/2020

- Advisor: Chirag J. Patel (Harvard Medical School)
- Repurposed a health insurance claims dataset of 45 million individuals to conduct the largest twin study in the United States (56,396 twin pairs), estimating genetic and environmental contributions to 560 phenotypes. Published in *Nature Genetics*; press coverage in the Washington Post, CNN, and The Verge.
- Developed the Exposome Data Warehouse, a database linking publicly available environmental data—socioeconomic, pollution, and climate—to electronic health records.
- NIH T32 Training Fellow, 2018–2020 (co-mentored by James Meigs, Massachusetts General Hospital).

Data Scientist, HelloWallet, Inc 04/2014 – 03/2015

- Designed and deployed a machine learning transaction classification system in a production consumer application; led migration of analytics infrastructure to a distributed computing environment.

Data Scientist, Zaloni, Inc 08/2013 – 04/2014

- Built the company's data science practice, delivering large-scale machine learning solutions for enterprise clients. Developed and taught an Introduction to Data Science course at several Fortune 500 companies.

Visiting Researcher (unaffiliated), SAMSI and Duke University 01/2012 – 08/2013

- Participated in SAMSI workshops and Mauro Maggioni's research group on geometric multi-resolution analysis and scalable manifold learning.

Postdoctoral Research Associate, North Carolina State University 08/2011 – 12/2011

- Advisor: Hamid Krim (North Carolina State University)
- Applied computational topology to data mining: persistent cohomology for nonlinear dimensionality reduction,

and Morse theory for community detection in networks.

Education

North Carolina State University, PhD in Mathematics 12/2010

- Thesis: The GIT Compactification of Quintic Threefolds
- Advisor: Amassa Fauntleroy

North Carolina State University, B.S. in Mathematics 12/2003

- Minor: Physics
- Magna Cum Laude

Additional Training: Statistical Genetics coursework with Alkes Price, Harvard T.H. Chan School of Public Health; Summer Institute in Statistical Genetics workshop with Peter Visscher.

Publications

Preprints Under Review

[1] Anjali Das, **Lakhani, Chirag M**, Vera M Mazeeva, Towfique Raj, and David A Knowles. "An empirical Bayes framework for burden and dispersion association tests helps prioritize rare variants associated with Alzheimer's disease". In: *medRxiv* (2026). URL: `https://doi.org/10.64898/2026.06.15.26354742`.

[2] Adithya Madduri, Randall Ellis, **Lakhani, Chirag M**, David Bennett, and Chirag J Patel. "Single-cell machine learning uncovers genetically anchored, cell-type specific programs of Alzheimer's disease". In: *medRxiv* (2026). URL: `https://doi.org/10.64898/2026.01.31.26345282`.

[3] **Lakhani, Chirag M**, Giacomo Cavalca, Anjing Liu, Rohan Nidumbur, Ru Feng, Towfique Raj, Philip De Jager, The Alzheimer's Disease Functional Genomics Consortium, Gao Wang, and David A Knowles. "Machine Learning-Based Prediction of Cell-type Resolved Brain eQTLs Enhances Discovery of Variants Explaining Alzheimer's Disease Heritability". In: *medRxiv* (2025). URL: `https://doi.org/10.64898/2025.12.03.25341562`.

[4] **Lakhani, Chirag M**, Jui-Shan T Lin, Anjali Das, Tatsuhiko Naito, Towfique Raj, and David A Knowles. "Integration of Deep Learning Annotations with Functional Genomics Improves Identification of Causal Alzheimer's Disease Variants". In: *medRxiv* (2025), pp. 2025–03. URL: `https://doi.org/10.1101/2025.03.07.25323578`.

[5] Anjing Liu, Roulan Jiang, Ruixi Li, Xuewei Cao, Zining Qi, Ru Feng, Hao Sun, Masashi Fujita, Natacha Comandante-Lou, **Lakhani, Chirag M**, Jenny Empawi, David A Knowles, Xiaoling Zhang, Kushal K Dey, Philip De Jager, David Bennett, The Alzheimer's Disease Functional Genomics Consortium, Tianying Wang, and Gao Wang. "Distributional genetic effects reveal context-dependent molecular regulation in human brain aging and Alzheimer's disease". In: *Research Square* (2025). URL: `https://doi.org/10.21203/rs.3.rs-8219833/v1`.

[6] Tatsuhiko Naito, Kosei Hirata, Beomjin Jang, **Lakhani, Chirag M**, Alice Buonfiglioli, Wan-Ping Lee, Otto Valladares, Li-San Wang, Yukinori Okada, Hong-Hee Won, et al. "Mosaic chromosomal alterations in blood are associated with an increased risk of Alzheimer's disease". In: *medRxiv* (2025), pp. 2025–05. URL: `https://doi.org/10.1101/2025.05.29.25328544`.

[7] Sara J Cromer, **Lakhani, Chirag M**, Deborah J Wexler, Sherri-Ann M Burnett-Bowie, Miriam Udler, and Chirag J Patel. "Geospatial analysis of individual and community-level socioeconomic factors impacting SARS-CoV-2 prevalence and outcomes". In: *MedRxiv* (2020). URL: `https://doi.org/10.1101/2020.09.30.20201830`.

Peer-Reviewed Publications

[8] Clara Yu, **Lakhani, Chirag M**, and David A Knowles. "Interpretable Deep Learning for Functional Fine-Mapping of Chromatin QTLs in Alzheimer's Disease". In: *2026 14th International Conference on Bioinformatics and Computational Biology (ICBCB)*. IEEE, 2026, pp. 85–89. URL: `https://doi.org/10.1109/ICBCB69424.2026.11621766`.

[9] Anjali Das, **Lakhani, Chirag M**, Chloé Terwagne, Jui-Shan T Lin, Tatsuhiko Naito, Towfique Raj, and David A Knowles. “Leveraging functional annotations to map rare variants associated with Alzheimer disease with gruyere”. In: *The American Journal of Human Genetics* 112.9 (2025), pp. 2138–2151. URL: `https://doi.org/10.1016/j.ajhg.2025.07.016`.

[10] Bryan Andrews, Chirayu Wongchokprasitti, Shyam Visweswaran, **Lakhani, Chirag M**, Chirag J Patel, and Gregory F Cooper. “A new method for estimating the probability of causal relationships from observational data: Application to the study of the short-term effects of air pollution on cardiovascular and respiratory disease”. In: *Artificial intelligence in medicine* 139 (2023), p. 102546. URL: `https://doi.org/10.1016/j.artmed.2023.102546`.

[11] Sara J Cromer, **Lakhani, Chirag M**, Josep M Mercader, Timothy D Majarian, Philip Schroeder, Joanne B Cole, Jose C Florez, Chirag J Patel, Alisa K Manning, Sherri-Ann M Burnett-Bowie, et al. “Association and interaction of genetics and area-level socioeconomic factors on the prevalence of type 2 diabetes and obesity”. In: *Diabetes Care* 46.5 (2023), pp. 944–952. URL: `https://doi.org/10.2337/dc22-1954`.

[12] Rockwell J Weiner, **Lakhani, Chirag M**, David A Knowles, and Gamze Gürsoy. “LDmat: efficiently queryable compression of linkage disequilibrium matrices”. In: *Bioinformatics* 39.2 (2023), btad092. URL: `https://doi.org/10.1093/bioinformatics/btad092`.

[13] Devi Sai Sri Kavya Boorgu, Shruthi Venkatesh, **Lakhani, Chirag M**, Elizabeth Walker, Ines M Aguerre, Claire Riley, Chirag J Patel, Philip L De Jager, and Zongqi Xia. “The impact of socioeconomic status on subsequent neurological outcomes in multiple sclerosis”. In: *Multiple sclerosis and related disorders* 65 (2022), p. 103994. URL: `https://doi.org/10.1016/j.msard.2022.103994`.

[14] Andrew Deonarine, Genevieve Lyons, **Lakhani, Chirag M**, and Walter De Brouwer. “Identifying communities at risk for COVID-19–related burden across 500 US cities and within New York City: unsupervised learning of the coprevalence of health indicators”. In: *JMIR Public Health and Surveillance* 7.8 (2021), e26604. URL: `https://doi.org/10.2196/26604`.

[15] Yixuan He, **Lakhani, Chirag M**, Danielle Rasooly, Arjun K Manrai, Ioanna Tzoulaki, and Chirag J Patel. “Comparisons of polyexposure, polygenic, and clinical risk scores in risk prediction of type 2 diabetes”. In: *Diabetes Care* 44.4 (2021), pp. 935–943. URL: `https://doi.org/10.2337/dc20-2049`.

[16] **Lakhani, Chirag M**, Braden T Tierney, Arjun K Manrai, Jian Yang, Peter M Visscher, and Chirag J Patel. “Repurposing large health insurance claims data to estimate genetic and environmental contributions in 560 phenotypes”. In: *Nature genetics* 51.2 (2019), pp. 327–334. URL: `https://doi.org/10.1038/s41588-018-0313-7`.

Earlier Work (Doctoral Training)

[17] **Lakhani, Chirag M**. “The GIT compactification of quintic threefolds”. In: *arXiv preprint arXiv:1010.3803* (2010). URL: `https://doi.org/10.48550/arXiv.1010.3803`.

Presentations

Conference Presentations

Machine Learning-Based Prediction of Cell-type Resolved Brain eQTLs Enhances Discovery of Variants Explaining Alzheimer’s Disease Heritability (Talk) — August 2026

NIH Alzheimer’s Disease Sequencing Project Annual Meeting, Washington, DC

Integration of Deep Learning Annotations with Functional Genomics Improves Identification of Causal Alzheimer’s Disease Variants (Poster) — May 2026

90th Cold Spring Harbor Symposium on Quantitative Biology: AI in Biology, Cold Spring Harbor, NY

Integration of Deep Learning Annotations with Functional Genomics Improves Identification of Causal Alzheimer’s Disease Variants (Poster) — July 2025

Gordon Research Conference on Alzheimer’s Disease, Ventura, CA

MLxQTL: Machine Learning Based Prediction of Single Cell eQTLs in Neuronal Cell Types (Talk) — May 2025

NIH Alzheimer’s Disease Sequencing Project xQTL Workshop, San Francisco, CA

Localization of Polygenic Signal in Alzheimer's Disease through the Integration of Cell-Type Functional Annotations and Deep Learning Models (Poster) — May 2024
Biology of Genomes, Cold Spring Harbor, NY

Deep Learning Approaches for Functional Genomic Variant Prediction using xQTL Data (Talk) — March 2024
NIH Alzheimer's Disease Sequencing Project xQTL Workshop, New York, NY

Localization of Polygenic Signal in Alzheimer's Disease through the Integration of Cell-Type Functional Annotations and Deep Learning Models (Poster) — March 2024
Broad Institute Variant-to-Function Meeting, Cambridge, MA

Localization of Polygenic Signal in Alzheimer's Disease through the Integration of Cell-Type Functional Annotations and Deep Learning Models (Poster) — October 2023
American Society of Human Genetics, Washington, DC

Repurposing Large Health Insurance Claims Data and Publicly Available Exposomic Data to Estimate Genetic and Environmental Contributions in 560 Phenotypes (Talk) — September 2019
Chan-Zuckerberg Initiative Workshop, CU Anschutz, Aurora, CO

Building a Search Engine to Find Environmental and Phenotypic Factors Associated with Health and Disease (Talk) — March 2017
NSF Big Data Spokes Meeting, Washington, DC

Systematic and Large-Scale Investigation of Twin and Sibling Concordance of 1723 Traits in a Nationally Representative Health Claims Cohort (Platform Talk) — October 2015
American Society of Human Genetics, Baltimore, MD

Consortium Presentations

Learning the Regulatory Code of Alzheimer's Disease Genomes (Talk) — October 2024
NIH Alzheimer's Disease Sequencing Project AI/ML External Advisory Board Meeting, Bethesda, MD

Learning the Regulatory Code of Alzheimer's Disease Genomes (Talk) — August 2024
NIH Alzheimer's Disease Sequencing Project AI/ML Working Group, Online

Deep Learning Analysis of Genetic Variants in PICALM/EED Locus (Talk) — March 2023
NIH Alzheimer's Disease Sequencing Project Functional Genomics Meeting, Online

Earlier Talks

Building an Exposome API — February 2016
Environmental Statistics Seminar, Harvard T.H. Chan School of Public Health, Boston, MA

Approximation of Kernel Matrices — November 2012
SAMSI Working Group, SAMSI, Research Triangle Park, NC

Tensor Decompositions in Signal Processing — October 2011
Tensor Seminar, North Carolina State University, Raleigh, NC

Fellowships, Grants, and Awards

Researcher, NIH/NIA R01 AG086467, *FunGen-xQTL: Unraveling the Genetic Basis of Molecular Functions in Alzheimer's Disease* (PI: Gao Wang, Columbia University) — 2026 – 2027

NIH T32 Training Fellowship, Harvard Medical School — 2018 – 2020

Stellar Abstract Award, Harvard Program in Quantitative Genomics Conference — November 2018

Microsoft Azure Research Award ($100,000 in computing credits) — May 2017

AMS Mathematical Research Community, Computational and Applied Topology June 2011

NSF Conference Travel Grants (MEGA 2011, Stockholm; Aspects of Moduli 2008, Pisa) 2008, 2011

Technical Skills and Software

Programming Languages: Python, R, C/C++, Java, Stan

Machine Learning: PyTorch, TensorFlow, Pyro, scikit-learn, Hugging Face; experiment management with Optuna and Weights & Biases

Genomics: genomic deep learning models (Enformer, ChromBPNet, DeepSEA); fine-mapping and heritability analysis (SuSiE, LDSC); whole-genome sequencing analysis at ADSP scale; single-cell genomics

Infrastructure: distributed computing (Spark, HPC/Slurm); cloud platforms (AWS, Azure, GCP); SQL and large-scale data engineering

Released Software: scEEMS, gruyere, parmigiano, LDmat (PyPI)